

SOLUTION

The Travelling Salesman Problem is solved by considering all possible routes and selecting the one with the minimum travel cost.

1. Represent the cities and the distances between them using a distance matrix.
2. Select a city as the starting point.
3. Generate all possible routes that visit every city exactly once.
4. Calculate the total distance for each route, including the return to the starting city.
5. Compare the distances of all possible routes.
6. Select the route with the minimum total distance.
7. Display the shortest path along with its total travel cost.

Note: For a small number of cities, all possible routes can be checked. For larger problems, optimization techniques and heuristic algorithms are preferred because the number of possible routes increases rapidly.